The Kadar–Yu root-distribution conjecture

Principia Math In Progress

22 September 2026

MathDB 372352. A proof of Morris–Martin Conjecture 4.5.1.

Task 1: all heights and all partitions

Fix an integer $l \geq -1$ and a partition $\lambda \vdash l+2$. Let $P_n^{(\lambda)}$ be the reduced monic polynomial sequence of [1, Theorem 3.19], defined by the one-cup Gram determinants and extended backwards by its Chebyshev recurrence. Put

$$N = l + 4, \qquad p_k(x) = P_{N+k}^{(\lambda)}(x), \qquad p_{k+1}(x) = xp_k(x) - p_{k-1}(x). \tag{1}$$

Here "reduced" means that consecutive polynomials are coprime. The minus sign in (1) is the determinant recurrence [1, (3.2.7)]; it also appears in [2, slide 17]. The isolated plus sign in [1, (3.1.1)] is inconsistent with these determinants and with its own classical examples.

Theorem 1. *For every such l, λ, each p_k, $k \geq -1$, has only simple real roots. The numbers of roots in $(-\infty, -2)$ and $(2, \infty)$ are separately nondecreasing for $k \geq 0$, bounded, and eventually constant. Moreover, the union of the roots of p_k, $k \geq 0$, that lie in $[-2, 2]$ is dense in that interval.*

We prove this using the established Gram-matrix structure, without assuming the explicit partition-family formulae conjectured elsewhere in [1].

A positive resolvent from one tail

Choose the real orthonormal Specht-module basis used in [1, Section 3.2]. At rank N, the Gram matrix has the form

$$G(x) = xI - A, \qquad A = A^T \in M_s(\mathbb{R}). \tag{2}$$

This is [1, (3.2.3)], with the constant matrix negated. Choose one terminal basis vector, corresponding to $u_{l+3,l+4}b_j$, and let $e \in \mathbb{R}^s$ be its unit coordinate vector. Extending just this tail by one vertex gives the partial Gram matrix

$$\begin{pmatrix} G(x) & e \\ e^T & x \end{pmatrix}. \tag{3}$$

The other tails remain at rank N. The proved one-tail factorization [1, Lemma 3.24 and (3.2.16)–(3.2.17)] says that the ratio of the determinant of (3) to $\det G(x)$ equals $p_1(x)/p_0(x)$. Its Schur complement therefore yields

$$\frac{p_1(x)}{p_0(x)} = x - e^T(xI - A)^{-1}e, \qquad \frac{p_{-1}(x)}{p_0(x)} = e^T(xI - A)^{-1}e. \tag{4}$$

These are identities of rational functions, so no invertibility assumption at particular real values is imposed.

By the spectral theorem, the last rational function is

$$m(x) = \sum_{j=1}^{q} \frac{w_j}{x - \theta_j}, \qquad \theta_1 < \cdots < \theta_q, \quad w_j > 0, \quad \sum_{j=1}^{q} w_j = 1. \tag{5}$$

Only eigenvalues with a nonzero projection of e are included. Thus (5) has exactly q distinct poles. Since p_0, p_1 are coprime, so are $p_0, p_{-1} = xp_0 - p_1$. Monicity and (4)–(5) identify

$$p_0(x) = \prod_{j=1}^{q} (x - \theta_j), \qquad \deg p_{-1} = q - 1, \tag{6}$$

with p_{-1} monic. In particular $q \geq 1$. On each interval (θ_j, θ_{j+1}),

$$m'(x) = -\sum_{i=1}^{q} \frac{w_i}{(x - \theta_i)^2} < 0.$$

The endpoint limits are $+\infty$ and $-\infty$. Consequently p_{-1} has one simple root in each of those $q-1$ intervals and no other roots. When $q = 1$, $p_{-1} = 1$, which is square free. This proves the otherwise exceptional index $k = -1$, as well as $k = 0$.

A finite matrix with an arbitrary path attached

Let

$$\mathcal{K} = \operatorname{span}\{e, Ae, A^2e, \ldots\}, \qquad H = A|_{\mathcal{K}}.$$

This is a reducing subspace for A. Its dimension is q, and e is cyclic for H. Indeed, its spectral projections are the q nonzero mutually orthogonal vectors in (5); polynomial interpolation in the distinct θ_j produces each projection from e.

Define H_k by attaching a path of k new vertices to e in H. Every new diagonal entry is zero and every path edge, including the attachment, has weight one. Thus $H_0 = H$ and $H_1 = \left(\begin{smallmatrix} H & e \\ e^T & 0 \end{smallmatrix}\right)$. Equation (4), followed by expansion at each new terminal vertex, gives

$$\det(xI - H_k) = p_k(x), \qquad k \geq 0. \tag{7}$$

For the initial step, $\det(xI - H) = p_0$ by (6), and $\det(xI - H_1) = p_0(x)(x - m(x)) = p_1(x)$. All subsequent determinants obey (1).

The terminal vector of the attached path is cyclic for H_k. Successive applications of H_k, subtracting already recovered coordinates, recover the whole path and then e. Continuing recovers every $H^j e$, hence the entire core $\mathcal{K}$. For $k = 0$, cyclicity was already established. A real symmetric matrix with a cyclic vector has simple spectrum: an eigenspace of dimension at least two would contain a nonzero vector perpendicular to the cyclic vector, and therefore to all its matrix iterates. Equation (7) now proves squarefreeness and reality for every $k \geq 0$.

Exterior roots and density

Let L_k and R_k count the eigenvalues of H_k strictly below -2 and strictly above 2, respectively. The matrix H_k is a principal submatrix of H_{k+1}. Cauchy interlacing consequently gives

$$L_{k+1} \geq L_k, \qquad R_{k+1} \geq R_k. \tag{8}$$

Deleting the q core coordinates from H_k leaves the ordinary k-vertex path matrix T_k. Its eigenvalues are

$$2\cos\frac{j\pi}{k+1}, \qquad 1 \leq j \leq k, \tag{9}$$

as follows by substituting the sine vectors into its three-term difference equation. They all lie in $(-2, 2)$. Interlacing for this deletion implies $L_k \leq q$ and $R_k \leq q$. Bounded nondecreasing integer sequences are eventually constant. This also accounts for possible roots exactly at the endpoints: the counts throughout are on the two open exterior intervals.

For completeness, interlacing gives a quantitative density result. Let $F_B(t)$ count the eigenvalues of B that are at most t. Deleting q coordinates yields

$$F_{T_k}(t) \leq F_{H_k}(t) \leq F_{T_k}(t) + q. \tag{10}$$

Define the arcsine distribution function

$$F(t) = \begin{cases} 0, & t \leq -2, \\ 1 - \pi^{-1}\arccos(t/2), & -2 < t < 2, \\ 1, & t \geq 2. \end{cases}$$

The explicit grid (9) shows $|F_{T_k}(t) - kF(t)| \leq 1$. Combining with (10), and using $\deg p_k = q + k$, gives

$$\sup_{t\in\mathbb{R}} \left| \frac{F_{H_k}(t)}{q+k} - F(t) \right| \leq \frac{q+1}{q+k}, \qquad k \geq 1. \tag{11}$$

Every nonempty open interval inside $(-2, 2)$ has positive arcsine measure. By (11), it contains roots of p_k for all sufficiently large k. Intervals arbitrarily near either endpoint give density in the closed interval $[-2, 2]$. Together with (6)–(8), this proves all clauses of the conjecture.

Tasks 2–5: proposed mathematical follow-ups

Task 2: Express the spectral points and weights in (5) directly from the Young diagram of λ, without constructing the full one-cup Gram matrix.

Task 3: Determine the eventual values of L_k, R_k and an explicit stabilization index in terms of l, λ.

Task 4: Determine the limits and convergence rates of the exterior roots, and characterize bounded generalized eigenvectors of the infinite path extension at -2 and 2.

Task 5: Derive a phase-shift formula for individual interior roots and sharp spacing estimates, refining the distribution bound (11).

References

[1] B. Morris and P. P. Martin, *On semisimplicity criteria and non-semisimple representation theory for the Kadar–Yu algebras*, arXiv:2512.24535v1 (31 December 2025), 66 pages. The Gram structure and one-tail factorization in Section 3 are established input; the target is Conjecture 4.5.1 on p. 35.

[2] B. Morris, *Interpolating between Temperley–Lieb and Brauer representation theory: generalised roots of unity*, ARTIN slides, Newcastle University, 11 September 2026.